20 Structured Nested If-Else Coding Questions in Python

A comprehensive workbook containing scenarios, algorithmic explanations, and precise solutions focusing strictly on hierarchical conditions.

1. Advanced E-Commerce Order Fraud Detection

Analyze high-risk transactions based on location variation, account lifetime, and absolute purchase values. Flag dangerous anomalies instantly.

```
home_country = "India"
tx_country = "USA"
account_age_days = 15
amount = 12000

if tx_country != home_country:
    if account_age_days < 30:
        if amount > 10000:
            print("Flagged status: Critical Fraud Risk! Transaction blocked.")
        else:
            print("Flagged status: Medium Risk. Require SMS OTP.")
    else:
        print("Flagged status: Low Risk. Standard international transaction.")
else:
    print("Flagged status: Safe. Local transaction.")
```

2. Corporate Employee Bonus Evaluation

Formulate corporate compensation grids assessing individual scorecards, professional tenure, and quantitative delivery quotas to calculate final stock and bonus weights.

```
performance_rating = 4.5
years_of_service = 6
projects_completed_pct = 95

if performance_rating >= 4.0:
    if years_of_service >= 5:
        if projects_completed_pct >= 90:
            print("Result: Eligible for Tier-1 Maximum Bonus + Stock Options.")
        else:
            print("Result: Eligible for Tier-2 Standard Bonus.")
    else:
        print("Result: Eligible for Tier-3 Accelerated Promotion Track.")
else:
    print("Result: Eligible for Standard Annual Increment only.")
```

3. Smart Home HVAC Climate Optimizer

Conserve electrical grids by cross-checking active motion trackers against internal thermal margins and relative humidity coefficients.

```
motion_detected = True
temperature_celsius = 29
humidity_pct = 75

if motion_detected:
    if temperature_celsius > 26:
        if humidity_pct > 70:
            print("HVAC Action: Activate AC in Dehumidifier/Dry Mode.")
        else:
            print("HVAC Action: Activate AC in Cool Mode at Maximum Fan Speed.")
    else:
        print("HVAC Action: Keep System in Fan-Only Mode to save power.")
else:
    print("HVAC Action: Put HVAC into Idle Eco-Mode.")
```

4. Institutional Loan Underwriting Filter

Assess fiscal risk profiles dynamically mapping independent metrics including credit tiers, active Debt-To-Income (DTI) metrics, and underlying structural collateral.

```
credit_score = 760
dti_ratio_pct = 38
has_high_collateral = True

if credit_score >= 750:
    if dti_ratio_pct > 35:
        if has_high_collateral:
            print("Underwriting: Approved with Premium Rate via Collateral Backin
g.")
        else:
            print("Underwriting: Rejected due to excessive debt metrics.")
    else:
        print("Underwriting: Approved immediately with Lowest Prime Rate.")
else:
    print("Underwriting: Rejected due to inadequate credit score.")
```

5. Automated Datacenter Firewall Rule Engine

Isolate malicious packet structures dynamically tracking incoming requests across unauthorized IP pools, system ports, and total data volumes.

```
is_whitelisted_ip = False
destination_port = 22
packet_size_kb = 4500

if not is_whitelisted_ip:
    if destination_port == 22:
        if packet_size_kb > 2000:
            print("Firewall Action: Drop Packet & Ban IP for DDoS Signature.")
```

```
        else:
            print("Firewall Action: Quarantine Request & Prompt for MFA.")
    else:
        print("Firewall Action: Allow traffic to Public Facing Web Ports.")
else:
    print("Firewall Action: Bypass all checks. Trusted Internal Node.")
```

6. Dynamic Airline Seat Upgrade Allocation

Route high-value elite flyers to luxury cabin inventory based on exact seat availability and current operational delay timestamps.

```
membership_tier = "Elite VIP"
business_seats_available = 0
flight_delay_minutes = 140

if membership_tier == "Elite VIP":
    if business_seats_available > 0:
        print("Upgrade Engine: Auto-Upgrade passenger to Business Class.")
    else:
        if flight_delay_minutes > 120:
            print("Upgrade Engine: Provide complimentary VIP Lounge Voucher.")
        else:
            print("Upgrade Engine: Place on Priority Standby List.")
else:
    print("Upgrade Engine: Keep Standard Economy seating assignment.")
```

7. 3D Graphics Engine Bounding-Box Collision Tester

Calculate spatial intersections in real-time within vector coordinate pipelines spanning structural X, Y, and Z-axes sequentially.

```
x_overlap = True
y_overlap = True
z_overlap = False

if x_overlap:
    if y_overlap:
        if z_overlap:
            print("Physics System: Multi-axis Collision Registered! Trigger Impac
t.")
        else:
            print("Physics System: Overlap from top-down view, heights differ.")
    else:
        print("Physics System: Separation identified on the Y-Axis.")
else:
    print("Physics System: No overlap. Objects are far apart.")
```

8. Algorithmic Cryptocurrency Trading Execution Engine

Execute microsecond market instructions tracing underlying macro trends, technical Relative Strength Index (RSI) limits, and trading volume spikes.

```
is_bullish_trend = True
rsi_value = 62
volume_surge_pct = 12

if is_bullish_trend:
    if rsi_value < 70:
        if volume_surge_pct > 10:
            print("Trading Action: Execute Market BUY Order immediately.")
        else:
            print("Trading Action: Hold Position. Volume is too low.")
    else:
        print("Trading Action: Do Not Buy. Asset is Overbought.")
else:
    print("Trading Action: Trend is Bearish. Keep funds safe in Stablecoins.")
```

9. Advanced Enterprise Password Policy Evaluator

Enforce deep structural validation frameworks assessing overall minimum lengths, inline digit elements, and special symbol arrays.

```
password_str = "SecurePass123"
has_min_length = len(password_str) >= 8
has_numbers = any(char.isdigit() for char in password_str)
has_special_chars = any(not char.isalnum() for char in password_str)

if has_min_length:
    if has_numbers:
        if has_special_chars:
            print("Password Strength: Excellent! Policy fully satisfied.")
        else:
            print("Password Strength: Moderate. Missing a special character.")
    else:
        print("Password Strength: Weak. Must include at least one numerical digit
.")
else:
    print("Password Strength: Rejected. Must be at least 8 characters long.")
```

10. Ride-Hailing Dispatch Fleet Optimizer

Coordinate physical driver dispatches dynamically analyzing relative distances, customer vehicle configurations, and live dynamic surge rates.

```
driver_distance_km = 3.2
driver_vehicle_type = "SUV"
requested_vehicle_type = "SUV"
driver_accepts_surge = False

if driver_distance_km <= 5.0:
    if driver_vehicle_type == requested_vehicle_type:
        if driver_accepts_surge:
            print("Dispatch Engine: Match Confirmed! Driver routed to customer.")
```

```
        else:
            print("Dispatch Engine: Driver rejected surge rate. Searching next...
")
    else:
        print("Dispatch Engine: Vehicle type mismatch. Driver bypassed.")
else:
    print("Dispatch Engine: Driver too far away. Search radius expanding.")
```

11. Multi-Tiered Warehouse Inventory Replenishment

Enforce automated inventory controls measuring safety stocks, pending purchase pipeline items, and immediate high-priority supplier backorders.

```
stock_level = 15
reorder_point = 20
incoming_shipment = False
lead_time_critical = True

if stock_level < reorder_point:
    if not incoming_shipment:
        if lead_time_critical:
            print("Inventory Action: Trigger Immediate Emergency Air-Freight Orde
r!")
        else:
            print("Inventory Action: Place Standard Replenishment Purchase Order.
")
    else:
        print("Inventory Action: Monitor stock. Shipment is already in transit.")
else:
    print("Inventory Action: Inventory levels optimal. No action required.")
```

12. Autonomous Vehicle Emergency Braking Framework

Process critical collision threat matrix checks scanning absolute lidar distances, structural traction maps, and cabin safety system statuses.

```
object_distance_m = 12
relative_speed_kmh = 65
wet_road_conditions = True

if object_distance_m < 20:
    if relative_speed_kmh > 40:
        if wet_road_conditions:
            print("Braking System: Trigger Maximum ABS and Emergency Sand Dispens
er!")
        else:
            print("Braking System: Initiate Standard Emergency Braking Sequence."
)
    else:
        print("Braking System: Apply Light Braking to maintain safe spacing.")
else:
    print("Braking System: Obstacle outside danger zone. Continue monitoring.")
```

13. Corporate Cloud Infrastructure Auto-Scaler

Govern server deployment matrix pools by calculating processor consumption metrics, historical inbound networks, and enterprise budgetary ceilings.

```
cpu_utilization_pct = 88
network_traffic_gbps = 8.5
budget_cap_reached = False

if cpu_utilization_pct > 80:
    if network_traffic_gbps > 5.0:
        if not budget_cap_reached:
            print("Infrastructure Action: Provision 5 New Compute Nodes Instantly
.")
        else:
            print("Infrastructure Action: Budget capped. Throttle non-essential t
asks.")
    else:
        print("Infrastructure Action: Optimize load balancing across current asse
ts.")
else:
    print("Infrastructure Action: System stable. Current node density sufficient.
")
```

14. Digital Streaming Royalty Distribution Parser

Distribute streaming micro-payments mapping platform tracking rules based on absolute track play lengths, geographic distribution nodes, and copyright tiers.

```
play_duration_sec = 45
is_premium_user = True
is_international_stream = True

if play_duration_sec >= 30:
    if is_premium_user:
        if is_international_stream:
            print("Royalty Calc: Allocate Tier-1 Premium International Payout Rat
e.")
        else:
            print("Royalty Calc: Allocate Tier-2 Premium Domestic Payout Rate.")
    else:
        print("Royalty Calc: Allocate Tier-3 Standard Ad-Supported Payout Rate.")
else:
    print("Royalty Calc: Stream duration too short. No royalty allocation compute
d.")
```

15. Medical Clinical Triage Assessment Matrix

Categorize emergency room patient triage states by processing cardiovascular vitals, neural scores, and age metrics.

```
systolic_bp = 85
gcs_score = 12
is_pediatric = False
```

```
if systolic_bp < 90:
    if gcs_score < 13:
        if is_pediatric:
            print("Triage Category: Immediate Level 1 Pediatric Resuscitation Req
uired.")
        else:
            print("Triage Category: Level 1 Adult Trauma Activation Required.")
    else:
        print("Triage Category: Level 2 Urgent Medical Intervention Indicated.")
else:
    print("Triage Category: Level 3 Standard Ward Assessment. Patient Stable.")
```

16. Smart Grid Electric Micro-Generation Router

Route localized green renewable energy inputs based on immediate localized capture levels, storage batteries, and variable market grid price levels.

```
solar_generation_kw = 8.5
battery_soc_pct = 92
grid_demand_high = True

if solar_generation_kw > 5.0:
    if battery_soc_pct > 90:
        if grid_demand_high:
            print("Grid Router: Sell Excess Renewable Generation to Main Grid Net
work.")
        else:
            print("Grid Router: Throttle Inverters to prevent battery overcharge.
")
    else:
        print("Grid Router: Route all solar energy to local storage batteries.")
else:
    print("Grid Router: Draw necessary operational power from localized battery c
ells.")
```

17. Fintech High-Frequency Account Liquidation Engine

Audit margin trade positions for risk liquidation using real-time debt-equity factors, margin balances, and underlying asset volatility trends.

```
margin_ratio_pct = 112
maintenance_margin_pct = 110
is_highly_volatile = True

if margin_ratio_pct < 115:
    if margin_ratio_pct <= maintenance_margin_pct:
        if is_highly_volatile:
            print("Liquidation Engine: Trigger Immediate Partial Market Liquidati
on!")
        else:
            print("Liquidation Engine: Issue Urgent Margin Call Notice to Account
.")
```

```
    else:
        print("Liquidation Engine: Account placed on high-frequency safety watch.
")
else:
    print("Liquidation Engine: Collateral healthy. No regulatory actions required
.")
```

18. Hotel Dynamic Room Booking Engine

Compute seasonal customer room rates by reviewing high-occupancy event calendars, loyalty histories, and booking lead-time windows.

```
is_peak_season = True
is_loyalty_member = True
days_until_checkin = 4

if is_peak_season:
    if not is_loyalty_member:
        if days_until_checkin < 7:
            print("Booking System: Apply Maximum Walk-In Pricing Multiplier (+40%
).")
        else:
            print("Booking System: Apply Standard High-Season Rate (+20%).")
    else:
        print("Booking System: Apply Premium Loyalty Flat Discount Rate.")
else:
    print("Booking System: Apply Promotional Off-Season Baseline Discount.")
```

19. Multi-Modal Logistics Transit Routing Selector

Choose shipping modalities by examining absolute item cargo metric ton weights, target delivery deadlines, and hazardous material codes.

```
cargo_weight_tons = 18.5
delivery_deadline_hours = 24
is_hazardous = True

if cargo_weight_tons > 15.0:
    if delivery_deadline_hours <= 48:
        if is_hazardous:
            print("Logistics Router: Deploy Dedicated Air-Express HazMat Vehicle
Fleet.")
        else:
            print("Logistics Router: Allocate Priority Heavy Freight Air Carrier
Split.")
    else:
        print("Logistics Router: Route via Standard Overland Rail Freight Network
.")
else:
    print("Logistics Router: Assign to Local Express Courier Delivery Network.")
```

20. Telecommunications Quality of Service (QoS) Broker

Sort live multimedia voice data packets over networks by parsing packet drop metrics, channel latency, and subscription tiers.

```
packet_loss_pct = 2.4
latency_ms = 115
subscriber_plan = "Platinum"

if packet_loss_pct > 1.5:
    if latency_ms > 100:
        if subscriber_plan == "Platinum":
            print("QoS Engine: Reroute traffic via High-Priority Premium Optical
Pipe.")
        else:
            print("QoS Engine: Downgrade stream quality to Audio-Only Mode.")
    else:
        print("QoS Engine: Increase buffering allocations to mitigate drops.")
else:
    print("QoS Engine: Connection parameters within nominal threshold values.")
```

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.